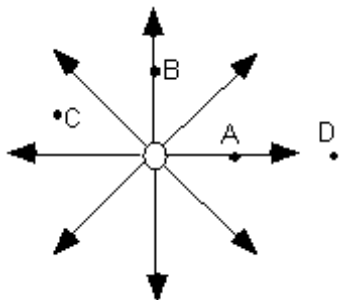


($k_e = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$, $m_e = 9.11 \times 10^{-31} \text{ kg}$, $e = 1.6 \times 10^{-19} \text{ C}$, $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N}\cdot\text{m}^2$, ($m_e = 9.1 \times 10^{-31} \text{ kg}$, $e = 1.6 \times 10^{-19} \text{ C}$)

1) **Exhibit 07-01**

1) _____

Refer to the electric field shown in the diagram below. It is produced by a single point charge at the center of the picture (Q)



Refer to Exhibit 07-01. What type of charge produces the electric field shown?

- A) neutral
C) positive

- B) impossible to tell
D) negative

2) Refer to Exhibit 07-01. If a new negative point charge is placed at location B directly above the one pictured, the force on it will

2) _____

- A) push it upward
C) be in an unknown direction.

- B) push it downward.
D) be zero

3) Refer to Exhibit 07-01. If a new positive point charge is placed at location A directly to the right of the one pictured, the force on it will

3) _____

- A) push it to the left.
C) be in an unknown direction

- B) be zero
D) push it to the right.

4) An ideal parallel-plate capacitor consists of two parallel plates of area A separated by a distance d . This capacitor is connected across a battery that maintains a constant potential difference between the plates. If the separation between the plates is now doubled, the magnitude of the charge on the plates will

4) _____

- A) quadruple.
B) be cut in fourth.
C) not change.
D) double.
E) be cut in half.

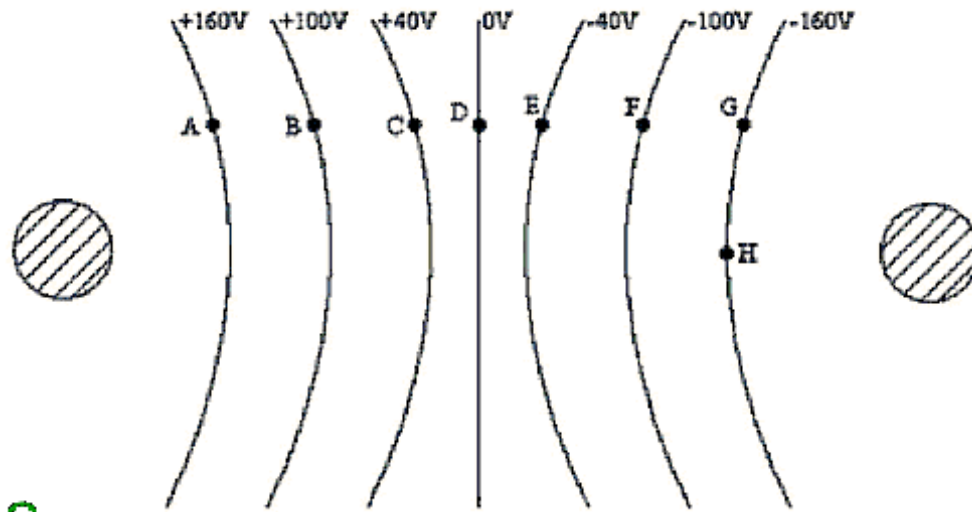
5) A 9.0 V battery moves 20 mC of charge through a circuit running from its positive terminal to its negative terminal. How much energy was delivered to the circuit?

5) _____

- A) $4.5 \times 10^3 \text{ J}$ B) 2.2 mJ C) 0.020 J D) 0.18 J

- 6) Two point charges, separated by 1.5 cm, have charge values of $+2.0\mu\text{C}$ and $-4.0\mu\text{C}$, respectively. What is the value of the mutual force between them? 6) _____
 A) $3.1 \times 10^{-3} \text{ N}$ B) $3.6 \times 10^{-8} \text{ N}$ C) $8.0 \times 10^{-12} \text{ N}$ D) 320 N
- 7) An electron is released from rest at the negative plate of a parallel-plate capacitor. If the distance across the plates is 5.0 mm and the potential difference across the plates is 5.0 V, with what velocity does the electron hit the positive plate? 7) _____
 A) $2.6 \times 10^5 \text{ m/s}$ B) $1.0 \times 10^6 \text{ m/s}$ C) $1.3 \times 10^6 \text{ m/s}$ D) $5.3 \times 10^6 \text{ m/s}$
- 8) If the size of the charge value is tripled for both of two point charges maintained at a constant separation, the mutual force between them will be changed by what factor? 8) _____
 A) 0.33 B) 3 C) 1/9 D) 9
- 9) A parallel-plate capacitor has dimensions 4.0 cm $\times$ 5.0 cm. The plates are separated by a 1.0-mm thickness of paper (dielectric constant $k = 3.7$). What is the charge that can be stored on this capacitor, when connected to a 1.5-V battery? 9) _____
 A) $20 \times 10^{-12} \text{ C}$ B) $4.8 \times 10^{-12} \text{ C}$ C) $4.8 \times 10^{-11} \text{ C}$ D) $9.8 \times 10^{-11} \text{ C}$
- 10) A Van Graaf generator has a potential of 5,000V. A Van Graaf generator is a metallic shell that you can charge. What is true 10) _____
 A) Inside the sphere, the electric field is 0 and the potential is 5000V
 B) Inside the sphere, both the electric field and the potential are constant (non zero)
 C) Inside the sphere, the electric field and the potential are 0
 D) Inside the sphere, the potential is 0V but the electric field is no
- 11) A parallel-plate capacitor has a capacitance of $20\mu\text{F}$. What potential difference across the plates is required to store $7.2 \times 10^{-4} \text{ C}$ on this capacitor? 11) _____
 A) $2.2 \times 10^{-2} \text{ V}$ B) 36 V C) 68 V D) $1.4 \times 10^{-8} \text{ V}$
- 12) Two ideal parallel-plate capacitors are identical in every respect except that one has twice the plate area of the other. If the smaller capacitor has capacitance C , the larger one has capacitance 12) _____
 A) $C/2$. B) C . C) $4C$. D) $2C$.
- 13) Two charges separated by one meter exert 1 - N forces on each other. If the charges are pulled 3 meters apart, the force on each charge will be 13) _____
 A) 0.33 N. B) 9 N. C) 0 N. D) 3 N. E) 0.11 N.
- 14) A repelling force must occur between two charged objects under which conditions? 14) _____
 A) Charges are of equal magnitude B) Charges are of unlike signs.
 C) Charges are of unequal magnitude. D) Charges are of like signs
- 15) In x-ray machines, electrons are subjected to electric fields as great as $6.0 \times 10^5 \text{ N/C}$. Find an electron's acceleration in this field. 15) _____
 A) $4.6 \times 10^{10} \text{ m/s}^2$ B) $3.6 \times 10^8 \text{ m/s}^2$
 C) $1.1 \times 10^{17} \text{ m/s}^2$ D) $5.4 \times 10^{13} \text{ m/s}^2$

- 16) Two point charges are 4 cm apart. They are moved to a new separation of 2 cm. By what factor does the resulting mutual force between them change? 16) _____
 A) 1/4 B) 2 C) 1/2 D) 4
- 17) An electron with a charge value of 1.6×10^{-19} C is moving in the presence of an electric field of 400 N/C. What force does the electron experience? 17) _____
 A) 4.9×10^{-17} N B) 2.3×10^{-22} N C) 1.9×10^{-21} N D) 6.4×10^{-17} N
- 18) The unit of electrical potential, the volt, is dimensionally equivalent to: 18) _____
 A) J×C. B) J/C. C) C/J D) F×C.
- 19) Find the electrical potential at 0.15 m from a point charge of 6.0 μC. 19) _____
 A) 5.4×10^4 V B) 3.6×10^5 V C) 2.4×10^6 V D) 1.2×10^7 V
- 20) The equipotential surfaces for two point charges are shown in the figure, with the value of potential marked on the line for each surface. What is the potential difference, $V_B - V_F$, between points B and F? 20) _____



- A) 0 B) 200V C) 100V D) -100V
- 21) Building on the previous questions. The electric field lines in the region are 21) _____
 A) always parallel to the equipotentials
 B) There is no field lines in the vicinity of equipotentials
 C) always perpendicular to the equipotentials
 D) Can be in any direction relative to equipotentials
- 22) Two point charges each have a value of 30.0 mC and are separated by a distance of 4.00 cm. What is the electric field midway between the two charges? ($k_e = 8.99 \times 10^9$ N·m²/C²) - Hint: draw the forces ! 22) _____
 A) 10.1×10^7 N/C B) 0 C) 40.5×10^7 N/C D) 20.3×10^7 N/C

$$|\vec{F}_E| = \frac{1}{4\pi\epsilon_0} \frac{|q_1 q_2|}{r^2}$$

$$\vec{E} = \frac{\vec{F}_E}{q}$$

$$|\vec{E}| = \frac{1}{4\pi\epsilon_0} \frac{|q|}{r^2}$$

$$\Delta U_E = q\Delta V$$

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

$$|\vec{E}| = \left| \frac{\Delta V}{\Delta r} \right|$$

$$\Delta V = \frac{Q}{C}$$

$$C = \kappa\epsilon_0 \frac{A}{d}$$

$$E = \frac{Q}{\epsilon_0 A}$$

$$U_C = \frac{1}{2} Q\Delta V = \frac{1}{2} C(\Delta V)^2$$

A	=	area
B	=	magnetic field
C	=	capacitance
d	=	distance
E	=	electric field
$\mathcal{E}$	=	emf
F	=	force
I	=	current
ℓ	=	length
P	=	power
Q	=	charge
q	=	point charge
R	=	resistance
r	=	separation
t	=	time
U	=	potential (stored) energy
V	=	electric potential
v	=	speed
κ	=	dielectric constant
ρ	=	resistivity
θ	=	angle
Φ	=	flux

23)

23) _____

Answer Key

Testname: TEST2_23

- 1) C
- 2) B
- 3) D
- 4) E
- 5) D
- 6) D
- 7) C
- 8) D
- 9) D
- 10) A
- 11) B
- 12) D
- 13) E
- 14) D
- 15) C
- 16) D
- 17) D
- 18) B
- 19) B
- 20) B
- 21) C
- 22) B
- 23) FALSE